

2.1 Water Cycle & Water Insecurity

Geography — Eduqas A-Level (A110QS) — Component 2: Global Systems

1. The Global Hydrological Cycle

The **hydrological cycle** is the continuous, closed-system circulation of water between the oceans, atmosphere, land and biosphere, powered by solar radiation and gravity. At the planetary scale it is a *closed system*: no significant water enters or leaves the Earth, so the total global volume of ~1.386 billion km³ is effectively constant. At local drainage-basin scale, however, the cycle operates as an *open system*: inputs (precipitation) and outputs (evapotranspiration, discharge) cross the basin boundary. The distinction between closed global and open local systems is a frequent AO1 requirement at Eduqas A-level.

Hydrological cycle: The continuous circulation of water between oceans, atmosphere, land surfaces and biosphere through the processes of evaporation, condensation, precipitation, infiltration, runoff and transpiration. Globally closed but locally open at basin scale.

Global Water Stores

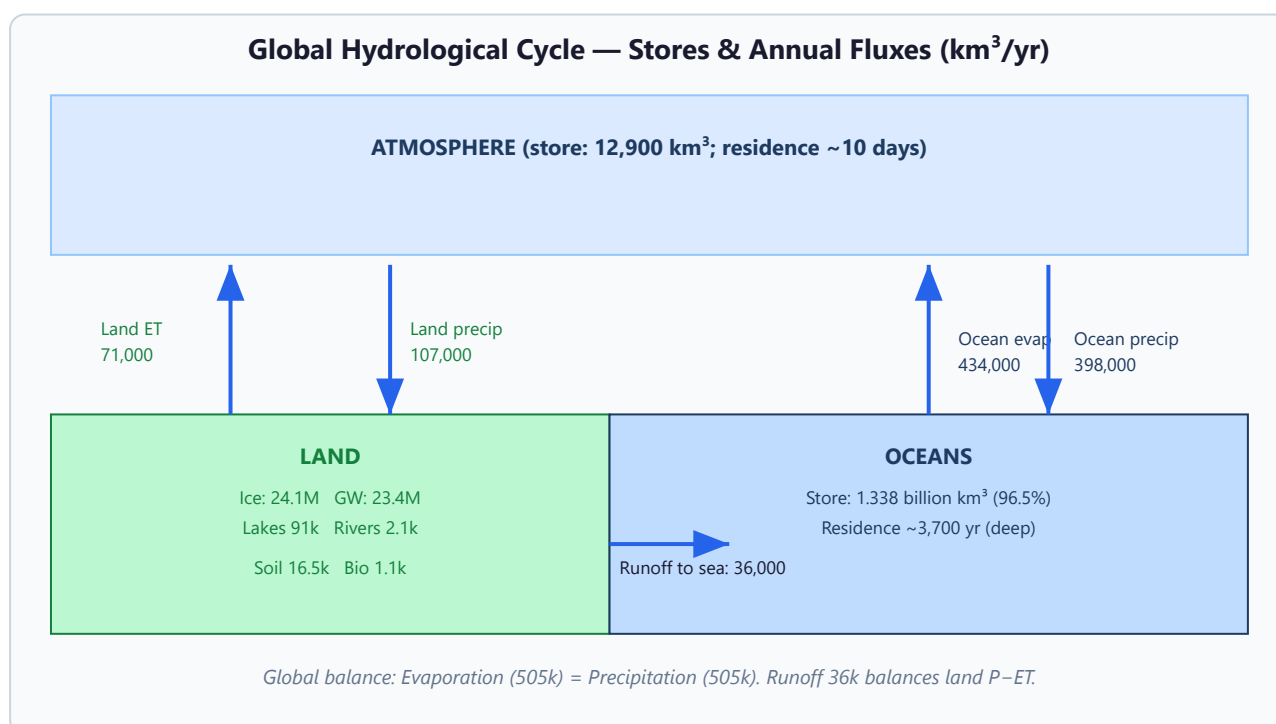
Although the hydrological cycle looks dynamic in textbook diagrams, 99.9% of global water sits almost motionless in three huge reservoirs: the oceans, the cryosphere (ice caps and glaciers) and groundwater. Only about 0.001% of global water (0.04% of fresh water) is in the atmosphere at any moment, yet this tiny flux delivers all terrestrial precipitation. Understanding the *asymmetry between store size and flux rate* is essential: small changes to a huge store (e.g. melting Greenland ice) can dwarf annual precipitation flux.

Store	Volume (km ³)	% of global water	% of fresh water
Oceans (saline)	1,338,000,000	96.5%	—
Ice caps, glaciers, permanent snow	24,064,000	1.74%	68.7%

Groundwater (total)	23,400,000	1.69%	—
of which fresh groundwater	10,530,000	0.76%	30.1%
Soil moisture	16,500	0.001%	0.05%
Lakes (fresh)	91,000	0.007%	0.26%
Atmospheric water vapour	12,900	0.001%	0.04%
Rivers	2,120	0.0002%	0.006%
Biological water (in organisms)	1,120	0.0001%	0.003%

Global Fluxes

The hydrological cycle's fluxes must balance annually. Global **evaporation** totals approximately 505,000 km³/yr, 86% of which (~434,000 km³) leaves the oceans; the remaining 14% (~71,000 km³) is continental evapotranspiration. **Precipitation** balances this: ~398,000 km³ falls on the oceans, ~107,000 km³ on the land. The imbalance between continental precipitation (107,000) and evapotranspiration (71,000) is ~36,000 km³/yr of **runoff** returning to the oceans via rivers and submarine groundwater discharge. This figure — global runoff — represents the *renewable freshwater resource* available to humanity.



Beyond evaporation and precipitation, several additional fluxes operate at surface and sub-surface scale: **transpiration** (water released by plant stomata; combined with direct evaporation gives *evapotranspiration*, ET); **infiltration** (water entering the soil at the surface); **percolation** (downward movement through unsaturated soil into groundwater); **throughflow** (lateral movement through soil to streams); **overland flow** (surface runoff); **groundwater flow**; and **channel flow** (river discharge). Each has a characteristic velocity and each feeds back onto stores at different timescales.

Exam tip — always quote a figure: In A-level questions on the hydrological cycle, adding precise data ("oceans hold 1.338 billion km³, or 96.5% of global water"; "global runoff ~36,000 km³/yr is the renewable freshwater resource") lifts answers from AO1 band 2 into band 3. Memorise at least three flux figures and three store percentages.

2. Residence Times and Management Implications

The **residence time** of a water store is the average length of time a water molecule spends there before moving on: $\text{Residence time} = \text{Store volume} / \text{Flux rate}$. Residence times vary across roughly six to seven orders of magnitude and directly control how *sustainable* human abstraction can be. Water drawn from a store with a short residence time (soil, rivers) is replenished within a year; water drawn from long-residence stores (deep aquifers, ice sheets) is effectively non-renewable on human timescales, making abstraction analogous to fossil-fuel mining.

Residence time: The average time a water molecule remains in a given store before passing through a flux to another store, calculated as store volume divided by flux rate. Short residence times imply rapid renewal; long residence times imply effectively non-renewable resources on human timescales.

Store	Approximate residence time	Management implication
Atmospheric water	~10 days	Rapid turnover; controls weather systems
Rivers	~2 weeks	Highly renewable; vulnerable to short droughts
Soil moisture	Weeks to months	Critical for agriculture; sensitive to land use

Wetlands	Months to years	Buffer floods; degraded by drainage
Lakes (freshwater)	~17 years (average)	Medium-term storage; sensitive to pollution
Shallow groundwater	10s to 100s years	Renewable only at slow rates
Deep (fossil) groundwater	1,000s to 10,000s years	Effectively non-renewable — “fossil water”
Mountain glaciers	20–100 years	Climate-sensitive buffer for Himalayan & Andean rivers
Ice sheets (Greenland, Antarctica)	10,000+ years	Long-term climate record; melt raises sea level
Deep ocean	~3,700 years	Thermohaline circulation drives global climate

Exam tip — link residence time to sustainability: Questions about the Ogallala aquifer (USA), the Great Man-Made River (Libya) or Saudi Arabia's abandoned wheat scheme all reward candidates who explain that pumping from a 10,000-year-residence aquifer is *mining fossil water*: abstraction exceeds recharge by orders of magnitude. Always distinguish renewable from non-renewable freshwater.

3. The Drainage Basin Hydrological Cycle

At the sub-global scale, water geographers work with the **drainage basin** as the fundamental unit of analysis. A drainage basin is the area of land drained by a single river and its tributaries, bounded by a topographic ridge called the *watershed*. Unlike the global cycle, a basin is an *open system*: precipitation crosses the watershed as input, while evapotranspiration (vertical) and river discharge (horizontal) cross it as outputs.

Drainage basin: The area of land drained by a river and its tributaries, bounded by a watershed. Functions as an open hydrological system with inputs of precipitation, stores in vegetation, surface and subsurface, flows between stores and outputs of evapotranspiration and channel discharge.

Inputs

- **Precipitation** — dominant input. The *type* (rain, snow, hail, sleet), *intensity* (mm/hr), *duration*, *spatial distribution* and *seasonality* all shape basin behaviour. Convective thunderstorms give short, high-intensity inputs that often exceed infiltration capacity; frontal rain gives prolonged lower-intensity inputs that saturate soils gradually; orographic rain concentrates precipitation on windward slopes.
- **Snowmelt** acts as a delayed precipitation input: snow falling in winter is stored until spring melt releases it rapidly, often generating annual peak discharges (e.g. Colorado River, Rhine).

Stores

- **Interception** — water held on the canopy, stems and litter of vegetation before reaching the ground. Interception loss can be 10–30% of gross rainfall for temperate deciduous woodland, 25–40% for coniferous forest.
- **Surface storage** — puddles (depression storage), ponds, lakes, wetlands.
- **Soil moisture** — water held in pore spaces in the soil above the water table. Field capacity is the maximum water the soil can hold against gravity.
- **Groundwater** — water held in saturated permeable rock below the water table. *Aquifers* are rock units capable of yielding usable quantities.
- **Channel storage** — water in rivers, streams and canals.

Aquifer: A body of saturated permeable rock (e.g. chalk, sandstone) capable of storing and transmitting groundwater in economically useful quantities. *Unconfined* aquifers have a free water table connected to the atmosphere; *confined* aquifers are sealed beneath an impermeable layer (aquiclude), often under artesian pressure.

Flows

- **Stemflow** — water running down tree trunks to the ground.
- **Throughfall** — water dripping through the canopy.
- **Infiltration** — vertical entry of water into the soil surface. *Infiltration capacity* is the maximum rate at which soil can accept water and declines during a storm as soil wets.
- **Percolation** — downward movement through unsaturated soil into bedrock and groundwater.
- **Throughflow** — lateral movement of water within soil, often along lines of greater permeability, toward streams. Typical velocities 1–100 mm/hr.

- **Overland flow** — surface runoff, of two kinds: *Hortonian (infiltration-excess) overland flow* occurs when rainfall intensity exceeds infiltration capacity (common in arid, compacted or urban surfaces); *saturation-excess overland flow* occurs when soil is already saturated and can accept no more water (common in temperate humid basins with rising water tables near channels).
- **Groundwater flow / baseflow** — slow flow of water through saturated rock, feeding rivers between storms and sustaining flow during droughts. Typical velocities 0.01–10 m/day.
- **Channel flow** — water flowing in rivers; measured as *discharge* ($Q = \text{velocity} \times \text{cross-sectional area}$, m^3/s).

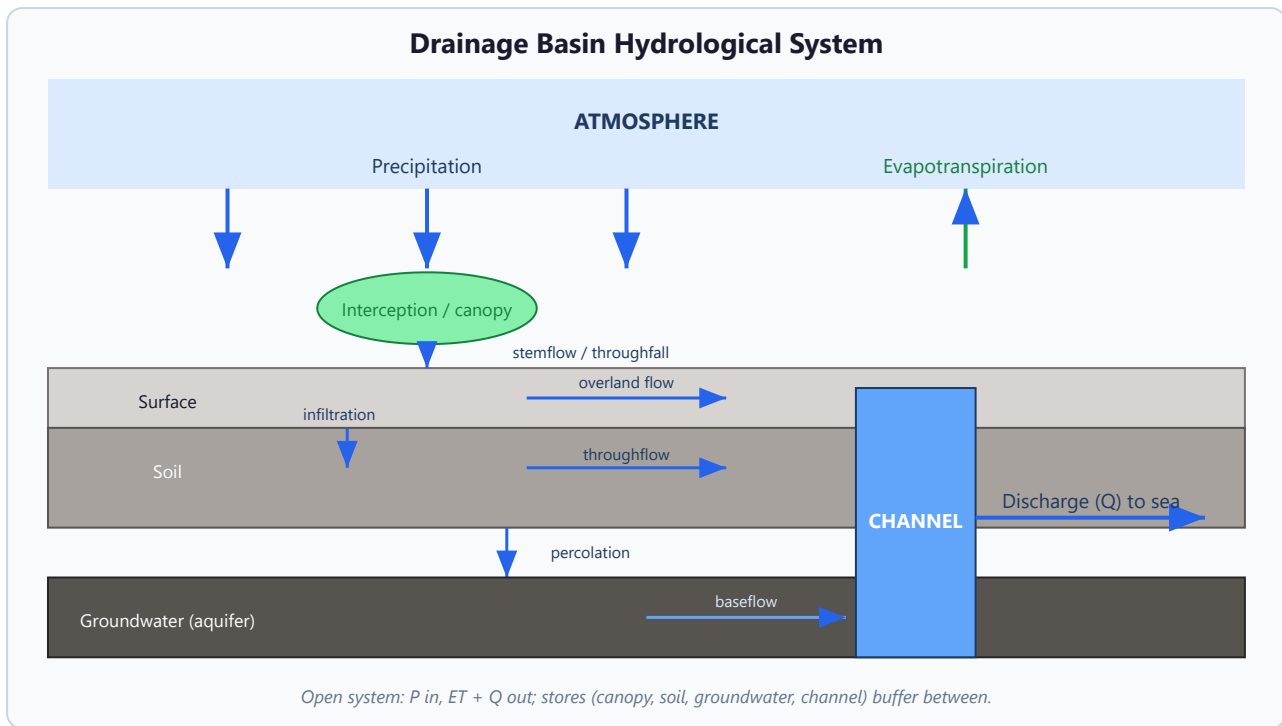
Infiltration: The vertical entry of water into the soil surface from rainfall or surface water. Rate controlled by soil texture, structure, moisture, vegetation, slope and rainfall intensity. Limited by the *infiltration capacity* of the soil.

Percolation: The slow downward movement of water through unsaturated soil and permeable rock under gravity, recharging groundwater stores. Distinct from infiltration (which is surface entry) and throughflow (which is lateral).

Throughflow: Lateral movement of water within the soil layer, downslope toward a stream channel, often concentrated in natural pipes and along the interface between soil horizons of different permeability. Velocities are intermediate between overland flow (fast) and groundwater flow (slow).

Outputs

- **Evapotranspiration (ET)** — combined evaporation from soil, water surfaces and vegetation plus plant transpiration. *Potential evapotranspiration* (PET) is the water that would be lost if unlimited water were available; *actual evapotranspiration* (AET) is the water actually lost, limited by available moisture.
- **Discharge to the sea or to a terminal basin** — the integrated basin output as river flow.



4. Water Budgets and Flood Hydrographs

The Water Budget Equation

For any basin over any time period, water balance requires inputs = outputs ± change in storage:

$$P = Q + E + \Delta S$$

where P is precipitation, Q is discharge (runoff), E is evapotranspiration and ΔS is change in basin storage (soil moisture + groundwater + surface water). On an annual timescale for a basin in steady state $\Delta S \sim 0$ and $P = Q + E$. Over shorter periods ΔS can be large and positive (winter recharge) or negative (summer depletion), producing seasonal mismatch between precipitation supply and available river flow.

Water budget (balance): The accounting equation for inputs, outputs and storage change in a drainage basin over a defined period: $P = Q + E + \Delta S$. Used by water managers to assess resource availability, identify seasonal deficits/surpluses and set sustainable abstraction limits.

Potential vs Actual Evapotranspiration

PET is set by atmospheric demand (solar radiation, temperature, humidity, wind); AET is constrained by moisture supply. In a UK temperate basin, PET during summer can exceed precipitation, creating a *soil moisture deficit*; AET is then less than PET because soil moisture is depleted. Autumn rainfall recharges soil to *field capacity*, and excess precipitation becomes *soil moisture surplus* that percolates to groundwater and sustains winter baseflow. Monthly water-balance graphs for Kew (southeast England) show deficit from ~May to September, utilisation in autumn, recharge in late autumn/winter, and surplus December to March.

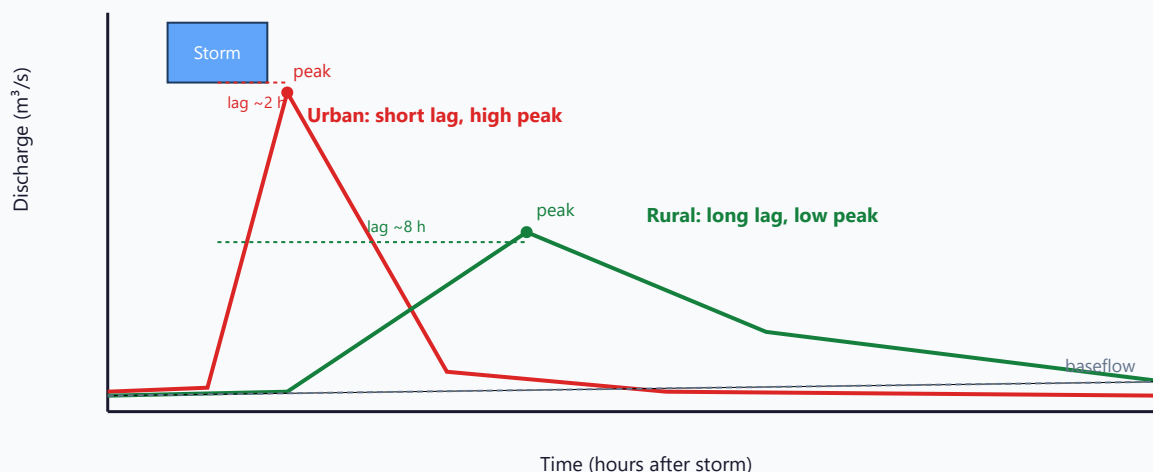
The Flood (Storm) Hydrograph

A **flood hydrograph** shows river discharge (m^3/s) against time following a rainfall event. Its shape records the way a basin translates rainfall into runoff. Key features:

- **Rising limb** — rapid increase in discharge as overland flow and throughflow reach the channel. Steepness reflects how quickly runoff is generated.
- **Peak discharge** — the maximum flow rate reached. Its magnitude controls flood severity.
- **Lag time** — the interval between peak rainfall and peak discharge. Short lag times characterise flashy basins (urban, impermeable); long lag times characterise permeable, vegetated basins.
- **Recessing (falling) limb** — gradual decline in discharge as direct runoff ceases and flow is sustained by slower throughflow and baseflow.
- **Baseflow** — the underlying flow sustained by groundwater between storms.
- **Storm flow (quickflow)** — the portion of discharge above baseflow, attributable to the storm event.

Lag time: The time interval between the peak of rainfall input and the peak of river discharge on a flood hydrograph. Short lag times (flashy response) indicate rapid runoff generation — typical of urban, small, steep, impermeable or saturated basins. Long lag times indicate buffered response — typical of large, vegetated, permeable basins.

Flood Hydrograph — Urbanised vs Rural Basin Response



Factors Affecting Hydrograph Shape

Factor	Flashy hydrograph (short lag, high peak)	Subdued hydrograph (long lag, low peak)
Basin size	Small (rapid transit of water to gauge)	Large (longer travel times; peaks attenuate)
Basin shape	Circular/fan-shaped (tributaries arrive simultaneously)	Elongated (tributaries arrive sequentially)
Relief	Steep (fast overland flow)	Gentle (slow overland flow)
Drainage density	High (many channels)	Low
Geology	Impermeable (clay, granite, urban tarmac)	Permeable (chalk, sandstone)
Vegetation	Sparse, bare soil, crops post-harvest	Dense woodland (interception + ET)
Antecedent soil moisture	Saturated (saturation-excess OLF)	Dry (high infiltration capacity)
Land use	Urbanised (tarmac, drains)	Rural, natural cover

Rainfall intensity / duration	Short intense storm (Hortonian OLF)	Prolonged low-intensity rain
Snowmelt	Rapid melt of accumulated snow	No snowpack

Exam tip — compare two named basins: Questions asking you to “analyse hydrograph shape” consistently reward contrasting two real basins. A good pair: the flashy River Roding (urban East London, peak 4 h after storm) versus the subdued River Frome (chalk, Dorset, peak 24–36 h after storm). Always link shape features to at least three causative factors.

5. Short- and Long-Term Variations in the Cycle

The hydrological cycle is not static. Natural climatic oscillations, human land-use change and long-term anthropogenic warming all drive variability at timescales from days to millennia.

Natural Variability

- **El Niño Southern Oscillation (ENSO)** — quasi-periodic (2–7 yr) coupled ocean–atmosphere oscillation in the tropical Pacific. In El Niño phases, weakened trade winds shift warm surface water east, causing drought in Australia, Indonesia and the Amazon and flooding along the Peru/Ecuador coast and south-east USA. La Niña reverses these anomalies. The 2015–16 El Niño, among the strongest on record, coincided with drought in Ethiopia, the worst on record in southern Africa, and severe bleaching of the Great Barrier Reef.
- **Indian (South-Asian) monsoon** — the seasonal reversal of wind direction and rainfall brings ~75% of India's annual precipitation between June and September. Inter-annual variability of $\pm 10\%$ of the mean delivers either widespread drought (1987, 2002, 2009, 2015) or flooding (2005 Mumbai, 2018 Kerala).
- **Sahel drought cycles** — the West African Sahel experienced catastrophic drought 1968–1985, linked to cooling of the North Atlantic (AMO), deforestation and overgrazing, with ~100,000 deaths. Partial recovery since the 1990s has been uneven and contested.
- **North Atlantic Oscillation (NAO)** — pressure seesaw between the Icelandic Low and Azores High, modulating winter storm tracks across the UK and western Europe.

Human-Induced Variability — Land Use Change

- **Deforestation** reduces interception and transpiration, lowers infiltration capacity via soil compaction and increases overland flow. In the Amazon, modelling suggests that removing 40% of forest cover could shift the basin past a tipping point where rainfall recycling fails and tropical forest converts to savanna.
- **Urbanisation** replaces permeable soil and vegetation with impermeable tarmac and concrete, removes interception, accelerates runoff via storm drains and increases flood peaks while reducing lag times. The Environment Agency's 2024 national flood risk assessment estimates that over 4 million properties in England are at risk of surface-water flooding.
- **Agriculture** — land drainage increases channel delivery; ploughing along slopes increases overland flow; compaction by heavy machinery reduces infiltration; irrigation returns water to atmosphere via ET.
- **Dams and reservoirs** store water, delay downstream discharge, reduce flood peaks, but also evaporate substantial volumes (Lake Nasser behind the Aswan High Dam loses ~10–15 km³/yr to evaporation) and trap sediment.

Climate Change and the Intensification of the Cycle

The Clausius–Clapeyron relationship means a warmer atmosphere holds ~7% more water vapour per 1°C of warming. The IPCC AR6 (2021) concludes with high confidence that the global hydrological cycle is intensifying: evaporation and precipitation totals both increase, extremes become more pronounced, and the spatial pattern shifts — wet regions tend to become wetter and dry regions drier (the “dry-get-drier, wet-get-wetter” paradigm, with important regional exceptions). Consequences include:

- More intense precipitation events, increasing flood risk even where annual totals are unchanged.
- Longer, more severe droughts in Mediterranean, south-west USA, southern Africa, Australia.
- Declining snowpack in mid-latitude mountains, earlier spring melt, and altered river seasonality (e.g. Sierra Nevada snowpack, California).
- Glacier retreat threatening “water tower” rivers — the Himalayan rivers (Indus, Ganges, Brahmaputra, Mekong, Yangtze) supply ~1.9 billion people.
- Sea-level rise and saline intrusion into coastal aquifers (e.g. Ghyben–Herzberg interface advance in Bangladesh, Florida, Nile Delta).

Exam tip — intensification, not simple “more water”: Top-band answers avoid vague statements such as “climate change means more rain”. Use the phrase *intensification of the hydrological cycle*,

link to the Clausius–Clapeyron 7%/°C rule, and specify both wet extremes (flooding) *and* dry extremes (drought, snowpack loss) in named regions.

6. Water Quality

Water security depends not only on *quantity* but also on *quality*. Water quality indicators span three domains: physical (turbidity, temperature, colour, taste, odour), chemical (pH, dissolved oxygen, nutrients, heavy metals, pesticides, chlorides, hardness) and biological (bacteria, viruses, protozoa, BOD — biochemical oxygen demand).

Pollution Sources

- **Industrial** — heavy metals (Cd, Hg, Pb), organic solvents, thermal pollution from power-station cooling water, hydrocarbons from refineries, acid mine drainage.
- **Agricultural** — nitrate and phosphate runoff from fertilisers, pesticides, organic loading from livestock slurry. Excess N and P cause *eutrophication*: algal blooms, oxygen depletion and fish kills in lakes, rivers and coastal zones (the Gulf of Mexico “dead zone” of ~15,000 km² is driven by Mississippi nutrient runoff).
- **Domestic sewage** — organic matter, pathogens (cholera, typhoid, cryptosporidium), phosphates from detergents, pharmaceuticals and micro-plastics. UK combined sewer overflows (CSOs) released raw sewage for over 3.6 million hours in 2023 (Environment Agency), triggering public outcry.
- **Mining** — acid mine drainage, sediment loading, heavy metals.
- **Plastic pollution** — ~1,000 rivers contribute ~80% of plastic entering oceans; the Pasig (Philippines), Citarum (Indonesia) and Yangtze (China) rank highest for plastic loading.

Eutrophication: The enrichment of water bodies with plant nutrients (principally nitrogen and phosphorus) from agricultural runoff, sewage or detergents, causing algal blooms, oxygen depletion through decomposition of dead algae, and mass mortality of aquatic life. A leading cause of freshwater ecological degradation globally.

7. Water Insecurity and Water Stress

Water security was defined by UN-Water (2013) as “the capacity of a population to safeguard sustainable access to adequate quantities of acceptable quality water for sustaining livelihoods, human well-being and socio-economic development, for ensuring protection against water-borne pollution and water-related disasters, and for preserving ecosystems in a climate of peace and political stability.” **Water insecurity** is therefore a multidimensional failure: insufficient quantity, inadequate quality, unreliable supply, or inaccessible cost/infrastructure.

Water stress: A situation in which renewable freshwater availability per capita falls below 1,700 m³/person/year (Falkenmark indicator). Stress becomes *water scarcity* below 1,000 m³, *absolute scarcity* below 500 m³. These thresholds are widely used by the UN and World Bank but mask important seasonal, spatial and quality variations.

Water Stress Thresholds (Falkenmark Indicator)

Category	Freshwater availability (m ³ /person/year)	Implication
Water sufficiency	> 1,700	Adequate for all sectors
Water stress	1,000–1,700	Periodic shortages; constraints on development
Water scarcity	500–1,000	Persistent shortages; limits agriculture and industry
Absolute scarcity	< 500	Severe constraints; dependence on imports / desalination

Physical vs Economic Water Scarcity

- **Physical (absolute) scarcity** — insufficient renewable water to meet demand regardless of wealth or infrastructure. Characteristic of arid and semi-arid regions: the Sahara, Arabian Peninsula, Central Asia, parts of the Colorado, Indus and Yellow basins. Approximately 1.2 billion people live in areas of physical water scarcity.

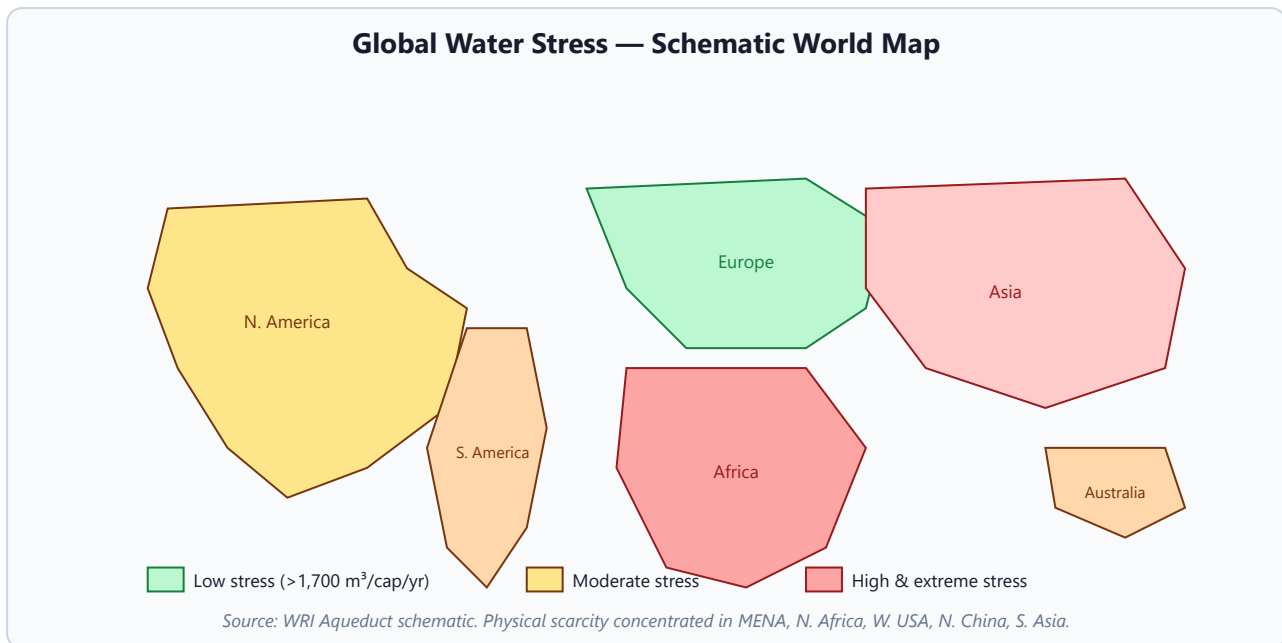
- **Economic water scarcity** — physical resources exist but are inaccessible because of lack of investment, infrastructure, institutions or governance. Characteristic of much of sub-Saharan Africa: the Congo basin has huge renewable water resources but over 60% of the population lacks safely managed drinking water. Approximately 1.6 billion people face economic water scarcity.

Drivers of Rising Water Demand

- **Population** — global demand is projected to rise ~55% by 2050 (OECD), principally in Asia and Africa.
- **Urbanisation** — 68% of humans will live in cities by 2050 (UN); urban per-capita demand exceeds rural.
- **Rising living standards** — dietary shift to meat and dairy sharply increases embedded water use (1 kg beef requires ~15,400 L water; 1 kg wheat ~1,800 L).
- **Irrigation** — ~70% of global freshwater withdrawal is for agriculture, dominated by irrigation; flood irrigation (often the cheapest) wastes up to 60% through evaporation and percolation.
- **Industrial use** — ~19% of global withdrawals; energy sector (cooling of thermoelectric power, hydraulic fracturing) competes directly with agriculture.

UN Sustainable Development Goal 6

SDG 6 ("Ensure availability and sustainable management of water and sanitation for all") targets universal access to safely managed drinking water and sanitation by 2030. Progress is off-track: the WHO/UNICEF Joint Monitoring Programme's 2025 report found that in 2024 2.1 billion people still lacked safely managed drinking water and 3.4 billion lacked safely managed sanitation. At recent rates universal access will not be reached until well after 2030.



8. Water Conflicts — Transboundary Basins

There are about **310 international (transboundary) river basins** (2018 update), covering ~47% of the Earth's land surface and home to over half of the world's population. Basins shared by multiple sovereign states generate intrinsic hydropolitics: upstream states enjoy first access and can divert, dam or pollute, while downstream states bear the consequences. The resulting *asymmetry of power* is exacerbated when upstream states are also richer or militarily stronger.

Two UN instruments govern shared freshwater:

- **UN Watercourses Convention (1997, in force 2014)** codifies principles of "equitable and reasonable utilisation" and "no significant harm". Fewer than 40 states have ratified; key basin powers (China, Turkey, Egypt) have not.
- **UNECE Water Convention (1992)**, originally European, opened to global accession 2016; promotes cooperation on transboundary basins.

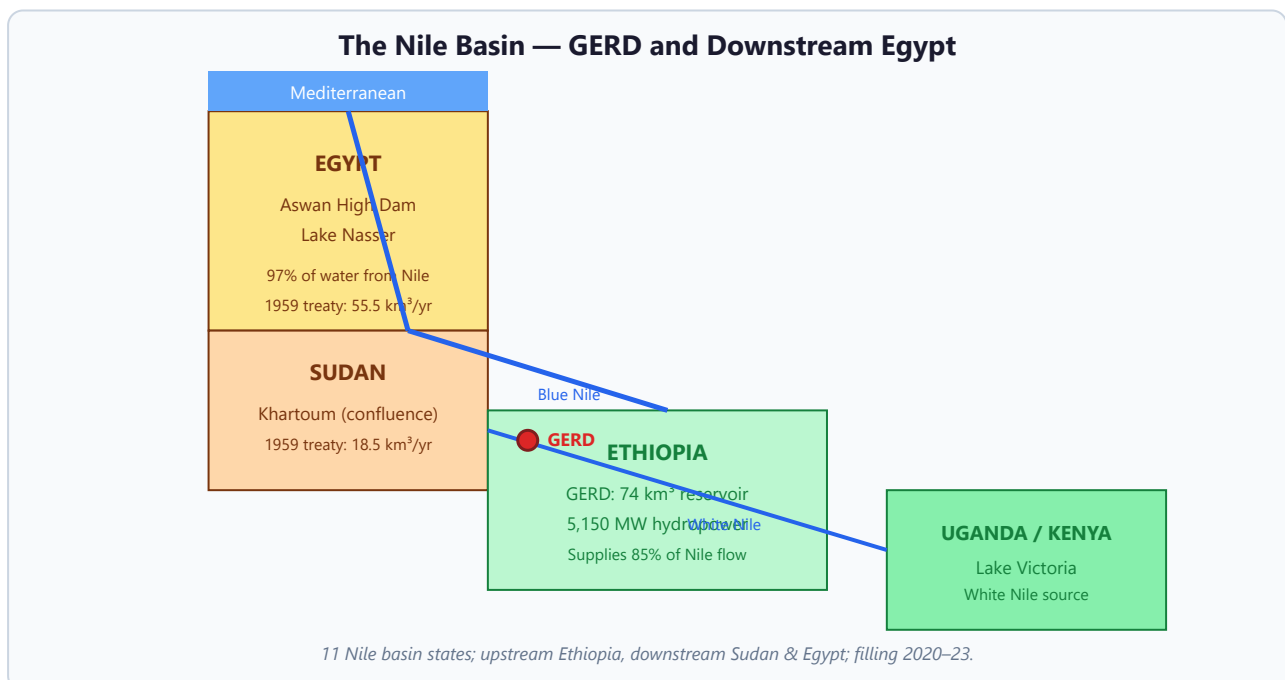
Despite scholarly pessimism about "water wars", the empirical record (Oregon State University's Transboundary Freshwater Dispute Database) records more instances of cooperation than conflict. However, GERD, the Jordan Basin, the Indus (India–Pakistan) and the Mekong all illustrate that latent tensions remain high.

9. Required Case Studies

Case study 1 — The Aral Sea disaster (Kazakhstan / Uzbekistan). Once the world's fourth-largest inland water body (68,000 km², 1960), the Aral Sea has lost ~90% of its area and ~75% of its volume. From the 1960s Soviet central planners diverted its two feeder rivers, the Amu Darya (south) and Syr Darya (north), to irrigate cotton and rice across the Kyzylkum and Karakum deserts, converting ~7 million hectares of drylands to cultivation. Combined withdrawals exceeded natural inflow; lake level fell ~23 m by 2010. Consequences include total fishery collapse (annual catch dropped from 48,000 tonnes to zero; the fishing port of Muynak now stands over 100 km from the shoreline); ecosystem collapse — loss of endemic species; emergence of the *Aralkum desert* on the exposed lake bed, from which salt-laden and pesticide-laden dust storms (~75 million tonnes/yr) degrade air quality and crop yields across Central Asia; major regional climate change, with colder winters and hotter summers; rising rates of respiratory illness, cancers and infant mortality in adjacent populations. Kazakhstan's 2005 *Kok-Aral Dam*, funded by the World Bank, partitioned off the North (Small) Aral Sea; surface level has since risen ~4 m and fish stocks have partially recovered. The much larger South (Large) Aral, within Uzbekistan, remains a shrinking hypersaline relic. The Aral case is the canonical example of unsustainable large-scale abstraction from a closed basin, and is frequently cited in Eduqas mark-schemes as a test of candidates' ability to link physical hydrology to political economy.

Case study 2 — Colorado River Basin, USA / Mexico. The Colorado drains 637,000 km² across seven US states (Wyoming, Colorado, Utah, New Mexico, Nevada, Arizona, California) plus Mexico, supplying ~40 million people and 2.2 million hectares of irrigated farmland (including California's Imperial Valley, which produces much of US winter vegetables). The **1922 Colorado River Compact** allocated 7.5 million acre-feet/yr (~9.3 km³) each to the Upper and Lower Basin states, with a further 1.5 MAF to Mexico (1944 treaty) — but was based on the wettest decade on record. Actual long-term mean flow is closer to 13–14 MAF, so the river is chronically *over-allocated*. Since 2000, the basin has experienced its worst megadrought in 1,200 years (tree-ring reconstructions, Williams et al. 2022); climate change has reduced flow by ~20% compared with the 20th-century mean. **Lake Mead**, the largest US reservoir, fell to 27% capacity in July 2022, following the first-ever federal *Tier 1 shortage declaration* (August 2021), which brought mandatory cuts to Arizona (18%) and Nevada (7%) allocations. Hoover Dam hydropower output has dropped ~33%. In May 2023, under Biden-administration pressure, Arizona, California and Nevada agreed to cut 3 MAF over 2023–2026 in exchange for \$1.2 bn federal funding. The Colorado is the textbook example of *over-allocation, climate-driven supply decline and institutional inertia* in an MEDC basin.

Case study 3 — Grand Ethiopian Renaissance Dam (GERD) and the Nile. The Nile (6,650 km) is shared by 11 states, with 85% of Egypt's flow originating from the Ethiopian highlands (Blue Nile) and the remainder from the White Nile (Lake Victoria). Colonial-era treaties (1929 Anglo-Egyptian, 1959 Egypt–Sudan) allocated $55.5 \text{ km}^3/\text{yr}$ to Egypt and $18.5 \text{ km}^3/\text{yr}$ to Sudan, giving Egypt an effective veto over upstream projects — and zero allocation to Ethiopia, despite it being the hydrological source. Construction of **GERD** (~\$5 bn, self-financed by Ethiopia) began in 2011 on the Blue Nile about 45 km from the Sudanese border. It is the largest hydroelectric plant in Africa: 5,150 MW installed capacity (scaled back from an original 6,450 MW design), a reservoir of 74 km^3 (roughly 1.5 years of Blue Nile flow). Ethiopia frames GERD as essential for lifting 60 million citizens out of energy poverty and for regional electricity export; Egypt views it as an existential threat to the ~97% of national water supply sourced from the Nile. The filling schedule became the focus of decade-long negotiations: Ethiopia filled the reservoir in stages 2020–2023 without a binding agreement, completing the fourth filling in September 2023 and a fifth in 2024; the dam was officially inaugurated in September 2025. Mediation by the African Union, USA and UAE has not produced a legally binding treaty on annual releases during droughts. Egypt has threatened military action; Sudan, initially opposed, shifted to accommodation. GERD illustrates *hydrological colonialism inverted*: upstream-downstream asymmetry, colonial-era treaty legacies, and whether cooperation can be institutionalised in a basin where ratification of the UN Watercourses Convention is refused.



Case study 4 — Jakarta water crisis and the capital relocation to Nusantara. Indonesia's capital Jakarta (~10.6 million population) faces an acute water and subsidence crisis. Only ~35–40% of residents receive piped water from the city utility (PAM Jaya); the remainder extract groundwater from shallow and deep aquifers via private wells. Combined with dense high-rise construction, this over-extraction has caused extreme **land subsidence**: rates reach 10–25 cm/year in the worst-affected northern districts (Muara Baru, Pluit), among the fastest in any major city on Earth. Approximately 40% of Jakarta is already below sea level; projections suggest a further expansion to ~95% of northern Jakarta below sea level by 2050 under combined subsidence and 20–25 cm sea-level rise. Tidal flooding is already frequent; the 2007 flood displaced 340,000 people; the 2020 New Year floods killed 67. Long-term engineering responses have included the stalled “Giant Sea Wall” megaproject, proposed in 2014. In August 2019 President Joko Widodo announced the relocation of the Indonesian capital to a new city, **Nusantara**, in East Kalimantan (Borneo). Planned to house 2 million by 2045, Nusantara construction began in 2022 and Independence Day was first celebrated there in August 2024, but the move has slowed under President Prabowo Subianto, and the formal transfer of capital status has been repeatedly delayed. Critics note that relocation addresses political symbolism but not the fundamental governance issue (lack of piped-water investment in Jakarta) and that Nusantara construction threatens Bornean rainforest and indigenous Dayak lands. Jakarta exemplifies *subsidence and governance failure* as drivers of urban water insecurity.

Case study 5 — Murray–Darling Basin, Australia. The Murray–Darling drains ~1.06 million km² (14% of Australia) across New South Wales, Victoria, Queensland, South Australia and the ACT, supporting AU\$24 bn of agricultural output (~40% of Australia's farm gate value) including rice, cotton, dairy and wine. Water extraction (largely for irrigation) grew from the 1920s to the 2000s to unsustainable levels: combined with the **Millennium Drought (2001–2009)** — one of Australia's worst on record — flows at the Murray Mouth fell to near zero, and Lake Alexandrina became hypersaline. Dryland salinisation affects ~17% of the basin due to rising water tables mobilising subsoil salt. In 2018–19, mass fish kills in the Darling near Menindee (estimated up to 1 million native Murray cod, golden perch and bony bream) became a political flashpoint, attributed to blue-green algal blooms, low flow and upstream cotton irrigation. The **Basin Plan 2012**, legislated under the Water Act 2007, set a target of returning 2,750 GL/yr from consumptive use to environmental flows, later raised to 3,200 GL. The plan has been contested by irrigators (who see it as an attack on rural livelihoods), by Indigenous First Nations (who argue their cultural water entitlements remain unrecognised despite the basin's Ngarrindjeri, Barkindji and Wiradjuri connections) and by the South Australian state government (which sees upstream compliance as inadequate). The Murray–Darling is

the archetypal MEDC over-allocation and *inter-state / Indigenous / environmental* conflict case, and is frequently paired with the Colorado in comparative Eduqas essays.

Case Study Summary Matrix

Case	Core driver	Key data	Main lesson
Aral Sea	Soviet cotton irrigation diversion	~90% area loss since 1960	Limits of large-scale abstraction
Colorado	Over-allocation + megadrought	Lake Mead 27% capacity 2022	MEDC supply-demand mismatch
GERD / Nile	Upstream–downstream asymmetry	74 km ³ reservoir; 11 states	Transboundary hydropolitics
Jakarta	Groundwater subsidence + SLR	25 cm/yr subsidence; 40% below sea level	Urban governance failure
Murray–Darling	Irrigation + drought + salinity	2012 Basin Plan: 3,200 GL return	Multi-stakeholder contested reform

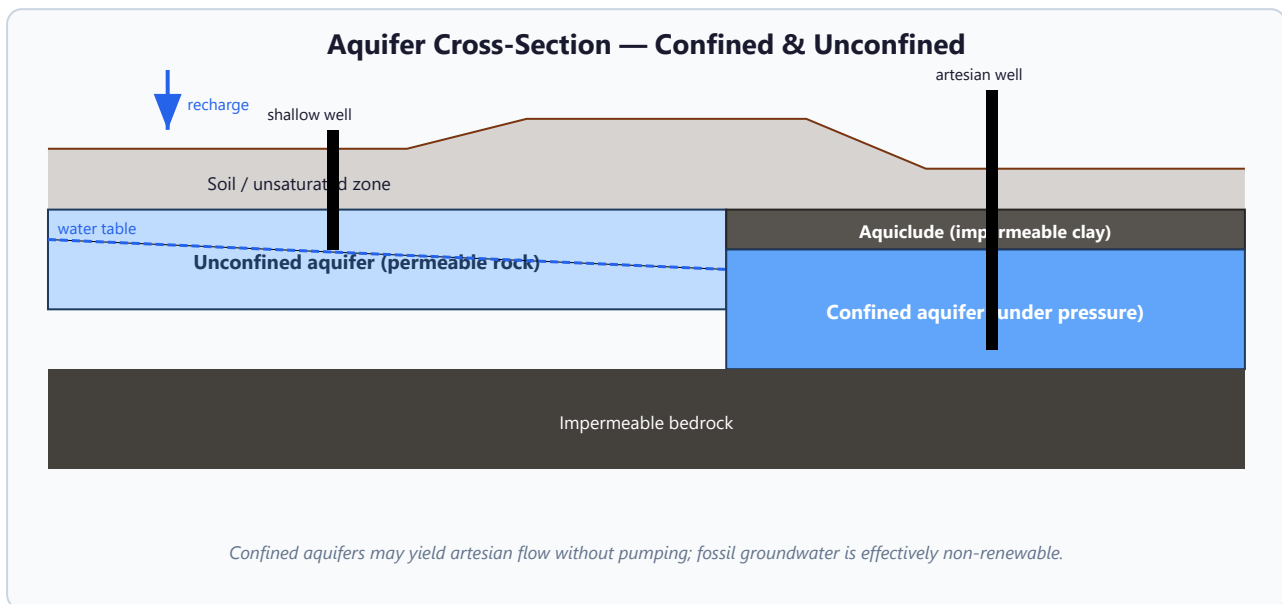
Exam tip — pair case studies thematically: For 20-mark extended responses, avoid listing all five cases. Instead pair them to frame an argument: Aral + Murray–Darling (over-abstraction in different political systems); Colorado + Nile (over-allocation and transboundary dispute); Jakarta alone (urban + climate). Always close with a point that synthesises physical with human geography.

10. Aquifers and Groundwater Management

Groundwater supplies roughly half of global domestic drinking water and ~43% of irrigation water. Two aquifer classes dominate:

- **Unconfined aquifers** have a free water table connected to the atmosphere via the unsaturated zone; recharge is rapid; pollution risk is high. Examples: UK Chalk aquifer (southern England), parts of the Ogallala (High Plains, USA).

- **Confined aquifers** are sealed below an impermeable *aquiclude*. Recharge occurs only at distant outcrops; water may be under pressure (*artesian*) and can flow to the surface without pumping. Examples: Great Artesian Basin (Australia), London Basin Chalk beneath the London Clay.



Unsustainable extraction causes: declining water tables (Ogallala has fallen 30+ m in parts of Kansas/Texas since 1950); saline intrusion in coastal aquifers (Gaza, Miami, Bangladesh); land subsidence (Jakarta, Mexico City has subsided ~10 m since 1900); loss of baseflow to rivers and wetlands; higher pumping energy costs. Managed Aquifer Recharge (MAR) schemes deliberately recharge aquifers with treated wastewater or diverted flood flows (e.g. Orange County, California's Groundwater Replenishment System can purify up to ~490,000 m³ (130 million gallons) a day for aquifer recharge).

11. Water Management Strategies

Hard Engineering

- **Large dams and reservoirs** — e.g. Three Gorges (China, 22,500 MW, 39 km³ storage), Aswan High Dam (Egypt), Hoover Dam. Provide hydropower, flood control, water supply and irrigation, but trap sediment, disrupt ecology, displace populations (Three Gorges displaced 1.3 million), and evaporate large volumes.
- **Large-scale water transfers** — China's **South–North Water Transfer Project** (SNWTP) moves up to 45 billion m³/yr from the water-rich Yangtze to water-scarce North China via three routes: Eastern (completed 2013, using the Grand Canal), Central (2014, Danjiangkou Reservoir to Beijing/Tianjin) and

Western (planned through the Tibetan plateau, delayed). Total cost ~\$80 bn. India's proposed National River Linking Project envisages similar transfers but is heavily contested.

- **Desalination** — reverse osmosis and multi-stage flash distillation. Global capacity ~110 million m³/day (~40 km³/yr). Dominant in the Gulf (Saudi Arabia, UAE, Qatar all over 50% of supply from desalination), Israel (~80% of domestic water from Mediterranean desalination at Sorek/Hadera/Ashkelon plants), Spain, Australia. Energy-intensive (~3–4 kWh/m³) and produces hypersaline brine that harms marine ecosystems.
- **Flood defences** — levees, embankments, storm-surge barriers (Thames Barrier, Netherlands Delta Works, Venice MOSE).

Soft / Integrated Management

- **Rainwater harvesting** — building-scale capture; Chennai, Tokyo, Berlin require it by code.
- **Water pricing & tariffs** — progressive tariffs (rising price per unit) fund infrastructure and discourage waste; Singapore's tariff rises steeply above 40 m³/month.
- **Leak reduction** — the UK loses ~20% of treated water to leaks (~3 bn L/day); acoustic monitoring and AI-guided repair programs target reduction.
- **Water-efficient technology** — low-flush toilets, smart meters, drip irrigation (saves 30–60% vs flood irrigation); Israel pioneered drip since 1960s.
- **Drought-tolerant crops** — CGIAR programmes, e.g. drought-tolerant maize for Africa, sorghum and millet revival.
- **Wastewater reuse** — Singapore's NEWater supplies ~40% of demand from treated wastewater (target 55% by 2060); Windhoek, Namibia has recycled sewage to drinking water since 1968.
- **Catchment management** — “working with nature” approaches such as reforestation, wetland restoration, leaky-dam flood alleviation (e.g. Pickering, North Yorkshire; Slowing the Flow scheme).

Integrated Water Resources Management (IWRM)

IWRM was formalised by the Global Water Partnership (Dublin Principles, 1992) and promoted by the UN via SDG 6.5. It is “a process which promotes the coordinated development and management of water, land and related resources in order to maximise economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems.” Key features: *basin-scale* planning units; *stakeholder participation* (farmers, industry, Indigenous, municipal); *balance* of supply and demand management; *integration* of water quantity, quality, ecology and economics. The EU Water Framework Directive (2000)

and South Africa's National Water Act (1998) are leading IWRM implementations; Murray–Darling Basin Plan is often cited as an IWRM attempt.

Integrated Water Resources Management (IWRM): A coordinated approach to managing water, land and related resources at the river-basin scale to balance economic, social and environmental objectives. Promoted by the UN via SDG 6.5; practical examples include the EU Water Framework Directive (2000) and the Murray–Darling Basin Plan (2012).

Virtual Water and Water Footprint

The concept of **virtual water** (Tony Allan, 1993, Stockholm Water Prize 2008) captures the volume of water embedded in the production of goods and services. A cup of coffee (125 mL) embeds ~140 L of water; a hamburger ~2,400 L; a cotton T-shirt ~2,700 L; 1 kg beef ~15,400 L; 1 kg rice ~2,500 L. International trade therefore embeds vast *virtual water flows*, enabling water-scarce states to import water-intensive products rather than produce them domestically (the *virtual water trade*). Critics argue virtual water masks the true social and ecological costs of exporting water from already-scarce regions (e.g. avocados exported from semi-arid Peru; almonds from drought-stressed California).

Virtual water: The volume of freshwater embedded in the production of a commodity, including the water consumed or polluted at each stage of the supply chain (Allan, 1993). A key concept because global trade flows of virtual water (~30% of global freshwater use) allow water-scarce states to secure food security without increasing domestic abstraction.

Hard vs Soft Strategies Compared

Dimension	Hard engineering	Soft / integrated
Scale	Large, centralised, capital-intensive	Often decentralised, cumulative
Time horizon	Rapid supply boost	Long-term demand management
Environmental cost	High (ecosystem disruption, sediment trapping)	Generally lower; sometimes net positive
Social cost	Displacement; inequitable benefits	Participatory; distributed benefits

Financial cost	Very high capital (tens of billions)	Low per project; cumulative
Climate resilience	Vulnerable to changing hydrology	Generally more adaptive
Examples	SNWTP, Three Gorges, GERD, desalination	IWRM, NEWater, rainwater harvesting, leak reduction, drip irrigation

Exam tip — always evaluate: “Assess” questions on management strategies require explicit evaluation, not description. Use the TEA frame: Technical feasibility, Economic cost-benefit, Acceptability (environmental, social, political). Top answers identify that hard and soft strategies are complementary, not alternatives (e.g. Israel combines desalination, drip irrigation and wastewater reuse).

12. Future Outlook

The 2020s and 2030s combine three interacting pressures: a global population rising from 8.1 bn (2024) to a projected 9.7 bn by 2050; continued urbanisation; and climate-driven intensification of hydrological variability. The OECD Environmental Outlook projects that water demand will grow ~55% by 2050, with manufacturing (+400%), thermal electricity generation (+140%) and domestic use (+130%) rising fastest. Already ~4 billion people — about two-thirds of the world — face severe water scarcity at least one month per year (Mekonnen & Hoekstra, 2016).

Water is increasingly reframed as a **security** issue. The Pacific Institute's Water Conflict Chronology (updated 2024) records a sharp rise in incidents of violence directly or indirectly linked to water since 2010, clustered in the Middle East, Sahel and South Asia. Intelligence agencies including the US Office of the Director of National Intelligence (DNI) now list water scarcity as a strategic risk factor for state fragility.

Technological and institutional responses converge on three axes:

- **Efficiency** — precision irrigation (soil-moisture sensors, satellite ET monitoring via MODIS/Landsat), AI leak detection, hydroponics and vertical farming for high-value crops.
- **New supply** — wastewater reuse (Windhoek, Singapore, Orange County), desalination at falling cost (Sorek-2, Israel, \$0.40/m³), and atmospheric water harvesting at small scale.

- **Governance reform** — transboundary institutions (Mekong River Commission, Senegal River Basin OMVS), rights-based approaches (2010 UN General Assembly resolution declaring a human right to water and sanitation), and recognising Indigenous cultural water rights (New Zealand's Whanganui River granted legal personhood, 2017).

The challenge is therefore *not* global water availability in aggregate — the planet has abundant freshwater — but its uneven distribution in space and time, its vulnerability to quality degradation, and the political economies that determine who gets it. Sustainable water futures will require behaviour change as much as engineering: dietary shift away from highly water-intensive meat and dairy, pricing reforms that signal scarcity, and political reforms that empower downstream, rural and Indigenous users whose voices have historically been marginalised.

Exam tip — structure 20-mark essays around physical-human integration: Questions at A-level almost always require synthesising physical hydrology with human agency. A strong structure is: (1) outline the physical process (cycle, residence time, hydrograph); (2) describe the human pressure (abstraction, pollution, climate); (3) present two contrasting case studies (one MEDC, one LEDC); (4) evaluate two management responses (one hard, one soft); (5) conclude with a judgement about sustainability. Explicitly naming Allan (virtual water), Falkenmark (stress thresholds), IWRM and SDG 6 signals engagement with the literature and reliably accesses the top AO1 band.

13. Extended Analysis: Selected Physical Processes

Precipitation Types — Generation Mechanisms

The physical mechanism of precipitation generation shapes its hydrograph impact. Three mechanisms dominate:

- **Convective rainfall** — intense surface heating triggers buoyant updrafts in unstable moist air; condensation above the lifting condensation level (LCL) releases latent heat, further destabilising the column. Produces cumulonimbus towers, localised high-intensity storms (often >50 mm/hr), hail and lightning. Characteristic of tropical afternoons, mid-latitude summer heatwaves, and the Indian monsoon. Generates Hortonian infiltration-excess overland flow readily, feeding flashy hydrographs.
- **Frontal (cyclonic) rainfall** — warm and cold air masses meet at a front; the warm, less dense air is lifted over cold air, cooled adiabatically and condensed. Produces prolonged (12–48 h), moderate-intensity rain along warm fronts, and shorter, more intense bursts at cold fronts. Characteristic of UK

and north-west European winters. Typically saturates soils progressively, generating saturation-excess overland flow from valley bottoms outwards (the “variable source area” model of Hewlett & Hibbert, 1967).

- **Orographic (relief) rainfall** — moist air forced upward over topography cools and condenses on windward slopes. Produces sharp rainfall gradients: Snowdonia west-side totals (>3,000 mm/yr) are four times east-side totals in the Vale of Clwyd. In tropical regions, the windward side of Hawaii receives >11,000 mm/yr while the leeward side receives <250 mm. The resultant *rain shadow* is a first-order control on basin water balance.

How raindrops form. Cloud droplets are far too small to fall, and two theories explain their growth to raindrop size. In the **collision–coalescence** process (dominant in warm tropical clouds), larger droplets fall faster, collide with smaller ones and absorb them. In the **Bergeron–Findeisen** process (dominant in cold mid-latitude clouds), ice crystals and supercooled water droplets coexist below 0 °C; because the saturation vapour pressure is lower over ice than over water, the ice crystals grow at the droplets' expense until they are heavy enough to fall, melting into rain if the air beneath is warm.

River regimes. A river regime is the annual pattern of discharge. *Simple* regimes have one period of high flow and one of low flow, driven by one dominant control (e.g. snow- and glacier-melt-fed Alpine rivers such as the upper Rhône, peaking in early summer). *Complex* regimes have two or more peaks because large basins draw on several climatic zones or inputs (e.g. the lower Rhine combines Alpine melt in summer with winter rainfall in its lower basin). Climate and season set the basic pattern; geology (permeable chalk sustains baseflow), vegetation and land use (urbanisation, reservoirs, abstraction) modify it.

Interception Losses — Canopy Hydrology

Interception loss has two components: *canopy storage* (capacity typically 0.5–2.5 mm for deciduous trees, 2–5 mm for conifers) and *canopy evaporation* during and after the storm. In low-intensity drizzle almost all rainfall may be intercepted; in heavy convective storms interception loss is proportionally smaller. Rutter-type canopy models partition gross rainfall into throughfall (drops falling through gaps or from leaf tips), stemflow (concentrated runoff down trunks, 1–10% of rainfall depending on bark roughness and canopy geometry), and interception evaporation. Empirical studies in Plynlimon (Wales) by the Institute of Hydrology showed that converting upland grassland to Sitka spruce plantation reduced annual streamflow by ~15–20% through increased interception and transpiration — evidence that fuelled debate over upland afforestation in water-supply catchments.

The Variable Source Area Concept

The classical Hortonian view that overland flow is widespread over a catchment is now known to be accurate only for arid, bare or urban surfaces. In humid temperate basins, Hewlett's *variable source area* (VSA) model shows that storm runoff is generated from a relatively small (~5–30%) dynamic contributing area near the stream channel, where soils are already near saturation. As the storm proceeds, the saturated area expands up-valley and onto footslopes, then contracts again after rain ends. VSA explains why basin-average infiltration capacities rarely predict observed hydrograph shape: only a fraction of the basin is hydrologically active at any moment. Modern distributed models (TOPMODEL, SHE) formalise this partial-area contribution.

Exam tip — go beyond Horton: Mentioning the Hewlett variable source area model (or TOPMODEL) alongside Hortonian overland flow shows awareness of how modern hydrology has refined classical concepts. Pair with a UK example (Plynlimon, River Severn headwaters) to secure AO2.

14. Extended Analysis: Selected Human Dimensions

Urbanisation and Sustainable Drainage (SuDS)

Urban surfaces convert permeable vegetated land into impermeable tarmac, concrete and rooftops, typically reducing infiltration to <10% of pre-urban rates while boosting runoff to >80% of rainfall. Storm drains then deliver this runoff to watercourses within minutes rather than days. Urban flood peaks can be 2–10× pre-urban values, with lag times shortened to 1–3 h; the phenomenon is visible in the flashy hydrographs of the River Quaggy (east London), River Roding, and Meadow Brook through urban Liverpool. In response, **Sustainable Drainage Systems (SuDS)** aim to restore natural hydrological function in cities via: permeable pavements, green roofs, swales, infiltration trenches, retention ponds, rain gardens and bioretention. The UK's 2010 Flood and Water Management Act included (Schedule 3) mandatory approval and adoption of SuDS for new developments; Wales brought this into force in January 2019, whereas implementation in England has been repeatedly delayed. Case study: Augustenborg, Malmö (Sweden), a post-1998-flood retrofit of a 1950s social-housing estate with green roofs, swales and open ditches, has reduced runoff by ~50% at sub-catchment scale.

Dam Removal — the Reverse Trend

While dam construction remains dominant globally, a counter-trend of strategic **dam removal** has emerged in MEDCs. The US has now removed over 2,000 obsolete or damaging dams (American Rivers,

2024). Major examples: removal of the Elwha (2011–14) and Glines Canyon (2011–14) dams in Washington State — the largest dam-removal project in history — has allowed salmon to return to 90 km of the Elwha river after a century's absence. The Klamath River dam removal project (California/Oregon, four dams removed 2023–24) is the largest contemporary example. In Europe, removal of the Vézins and La Roche-qui-Boit dams on the Sélune (Normandy, France, 2019–2022) has similar aims. Dam removal is justified by: declining benefit-cost ratios (many dams are aged/unprofitable), ecological restoration (anadromous fish), flood-risk reduction by restoring floodplain capacity, and safety (ageing dam infrastructure). In the UK, weir removals and fish passes (e.g. the “Unlocking the Severn” project) are part of the same trend.

Water Pricing Politics

Water pricing is politically charged because water is simultaneously a basic human right (UN 2010 resolution) and an economic good (Dublin Principles, 1992). Under-pricing leads to waste and under-investment; over-pricing excludes the poor. Approaches include:

- **Progressive (block) tariffs** — low or free for basic needs, then rising steeply. South Africa guarantees 6,000 L/household/month free, then charges above (Free Basic Water Policy, 2001).
- **Flat tariffs** — common in state-socialist legacies but inefficient; ~20% of water in many ex-Soviet cities is unmetered.
- **Full cost recovery** — as advocated by the World Bank; controversial where implemented (Cochabamba, Bolivia “Water War” 2000 overturned privatisation after mass protest).

Cochabamba remains a cautionary reference: Bechtel subsidiary Aguas del Tunari raised water prices by 35–50% in early 2000, triggering mass protests in which one student was killed; the concession was cancelled and the company sued Bolivia for \$50 million. Economists and geographers cite it to caution against uncritical privatisation of basic resources.

Exam tip — use the UN Human Right to Water: In any question involving water access inequality, citing UN General Assembly Resolution 64/292 (2010) declaring “the right to safe and clean drinking water and sanitation as a human right” lifts answers into AO1 top bands and provides an evaluative frame against which to judge pricing, privatisation and SDG 6 performance.

15. Synoptic Links — Water in Other Eduqas Topics

Water security is synoptic by design. Strong A-level answers connect it explicitly to other parts of the A110QS specification:

- **Component 2: Global Governance** — SDG 6, transboundary treaties (Nile, Indus Waters Treaty 1960), UN Watercourses Convention, human right to water.
- **Component 1: Coastal Landscapes** — sea-level rise and saline aquifer intrusion (Bangladesh, Nile Delta, Florida Everglades); fluvial sediment starvation behind dams leading to coastal erosion (Nile Delta subsidence).
- **Component 1: Changing Places** — place meaning attached to rivers (Ganges, Whanganui); cultural water in Indigenous geographies; water as local heritage (Welsh cwm streams).
- **Component 3: Tectonic Hazards** — dam-induced seismicity (Koyna, India; Zipingpu suspected trigger of 2008 Sichuan earthquake debate).
- **Independent Investigation (NEA)** — water quality, SuDS effectiveness, hydrograph comparison and flood-risk perception are all standard NEA research topics.

16. Further Case Study Material — Supplementary Examples

Himalayan “Water Towers”

The Hindu Kush–Himalayan (HKH) region contains the greatest concentration of ice outside the polar regions, feeding ten major river systems (Indus, Ganges, Brahmaputra, Yangtze, Yellow, Mekong, Salween, Irrawaddy, Amu Darya, Tarim) that support ~1.9 billion people across Afghanistan, Pakistan, India, Nepal, Bhutan, Bangladesh, China and Southeast Asia. The 2019 ICIMOD Hindu Kush Himalaya Assessment concluded that even meeting the Paris Agreement 1.5°C target would lose ~33% of HKH glacier ice by 2100; unchecked warming would lose ~66%. The short-term effect is *increased* summer runoff from melt; the long-term effect is collapse of dry-season baseflow. For the Indus (Pakistan), which receives ~50% of dry-season flow from melt, the consequences compound existing tensions with India over the 1960 Indus Waters Treaty and with Afghan upstream hydropower proposals. For Bangladesh, changing Ganges/Brahmaputra flow interacts with sea-level rise and cyclone intensification to threaten the Sundarbans and the 170-million-strong delta population.

California Drought Cycles

California's 2012–2016 drought was classified the worst in 1,200 years on tree-ring evidence; the state mandated a 25% urban water-use cut in 2015 (first such order in state history). Sierra Nevada snowpack —

which delivers ~30% of California's water — fell to 5% of average in April 2015. Agriculture in the Central Valley compensated by over-pumping groundwater, causing subsidence of up to 0.6 m/yr in parts of the San Joaquin Valley (Farr et al., USGS). The 2014 Sustainable Groundwater Management Act (SGMA) for the first time required local agencies to reach “sustainable yield” by 2040, but implementation has been contested by almond and alfalfa growers whose water-intensive exports are heavily dependent on groundwater mining. A second drought 2020–2022 brought record-low reservoirs before a “whiplash” wet winter 2022–23 delivered atmospheric-river-driven floods, exemplifying climate-driven intensification of both extremes.

Cape Town “Day Zero” 2018

Between 2015 and 2018, a multi-year drought brought Cape Town within weeks of *Day Zero* — the forecast day when municipal taps would be cut off and residents would queue for 25 L rations at collection points. The city's storage (Theewaterskloof Dam and five others) fell from 79% in September 2015 to ~13% in February 2018. Emergency measures included: an 87 L/person/day limit (later 50 L); progressive tariffs rising to over ten times normal levels above 10.5 kL/month; a public “water map” naming over-consuming households; restrictions on swimming pools, car washing and garden irrigation. Per-capita consumption fell from ~540 L/day (2015) to ~280 L (early 2018) — a halving within months. Winter 2018 rainfall arrived in time; Day Zero was postponed and then cancelled. The case is widely cited as evidence that rapid, coordinated demand management can avert acute crisis without requiring hard engineering, and it transformed Cape Town's long-term water strategy toward aquifer recharge, desalination (small-scale) and extensive reuse.

Great Man-Made River, Libya

Completed in 2007 at a cost of ~\$25 bn, the Great Man-Made River (GMMR) pumps ~6.5 million m³/day of fossil groundwater from the Nubian Sandstone Aquifer System (NSAS) across the Sahara to Libyan coastal cities (Tripoli, Benghazi, Sirte). The NSAS, recharged during the Pleistocene pluvial periods over 10,000 years ago, holds an estimated 150,000 km³ of water — but at current abstraction rates will be depleted in 60–100 years. Libya has no meaningful replenishment; the scheme is literal *fossil water mining*. The 2011 NATO intervention partially damaged GMMR infrastructure; the subsequent Libyan civil war has left maintenance inconsistent. GMMR exemplifies the short-term appeal and long-term unsustainability of non-renewable water use.

Three Gorges Dam, China

Completed 2006 (hydropower operational 2012), the Three Gorges Dam on the Yangtze is the world's largest hydroelectric plant by installed capacity: 22,500 MW across 32 turbines. Its reservoir (39.3 km³) extends 600 km upstream to Chongqing. Benefits claimed: ~95 TWh/yr of carbon-free electricity (equivalent to ~50 million tonnes of CO₂ displaced per year if replacing coal); flood protection for ~15 million people downstream (notably during the 2010 and 2020 floods); improved Yangtze navigation, allowing 10,000-tonne ships to reach Chongqing. Costs: displacement of ~1.3 million people; submergence of archaeological sites; loss of endemic species (the Yangtze River dolphin was declared functionally extinct in 2007); sediment starvation downstream, contributing to erosion of the Yangtze Delta and Shanghai coastline; concerns over induced seismicity; water quality decline in the reservoir due to agricultural and industrial run-off; and high evaporative losses.

Additional case study — Singapore's "Four National Taps". Singapore, a small island-state with no natural freshwater resources beyond rainfall, has built a model water security strategy around four complementary sources. (1) *Local catchment water*: 17 reservoirs catching rainfall over two-thirds of Singapore's surface area; the Marina Barrage (2008) impounds the Marina Reservoir in the heart of the city. (2) *Imported water*: under bilateral treaties with Malaysia (1961 and 1962), Singapore draws up to 250 million gallons/day from Johor; the 1962 agreement expires in 2061. (3) *NEWater*: treated wastewater reclaimed to ultra-pure standard by microfiltration, reverse osmosis and UV disinfection, launched 2003; now supplies ~40% of demand, projected to reach 55% by 2060. (4) *Desalination*: five plants (SingSpring 2005, Tuaspring 2013, Tuas 2018, Keppel Marina East 2020, Jurong Island 2022) supplying ~25% of demand. Complemented by the Active, Beautiful, Clean (ABC) Waters programme that converts concrete drainage channels into naturalised stream corridors (Bishan-Ang Mo Kio Park, 2012). Singapore's combined strategy illustrates IWRM implemented at city-state scale and is widely referenced as a benchmark for Gulf, Mediterranean and small island states.

Water, Gender and Health

In LEDC rural contexts, water collection burdens fall disproportionately on women and girls. UN Women estimates that globally women and girls spend a combined 200 million hours/day collecting water, with median one-way trip distances of 6 km in sub-Saharan Africa. This labour imposes: opportunity costs on schooling (girls often withdraw from secondary school); musculoskeletal injury from carrying 20–25 L head loads; and sexual violence risk at remote collection points. Water insecurity is therefore also a gender-inequality issue. WaterAid and similar NGOs emphasise that investing in local piped supply (not only distant dams) yields disproportionate gender, education and health benefits. SDG 6.1 targets universal equitable access — monitoring shows gaps widest for women in rural Africa.

Water and the Energy–Food–Water Nexus

Water, energy and food are deeply coupled: thermoelectric generation requires cooling water (~40% of US freshwater withdrawals); water supply, treatment and distribution require energy (~3–5% of national electricity in many MEDCs); food production consumes ~70% of freshwater withdrawals. The *nexus* framing (Bonn 2011 conference) warns that single-sector policies produce unintended consequences: subsidising electricity for irrigation pumps in Punjab accelerated groundwater depletion; expanding hydropower can reduce downstream agricultural water; biofuel mandates can divert food-crop water to fuel crops. Nexus thinking reinforces the argument for IWRM and cross-sectoral governance.

17. Quantitative Reference Data

Store Volumes and Fluxes — Detailed Figures

- Total Earth water: ~1,386,000,000 km³
- Ocean water: 1,338,000,000 km³ (96.54%)
- Ice caps and glaciers: 24,064,000 km³ (1.74%)
- Antarctic ice sheet: ~26,500,000 km³ (90% of global land ice)
- Greenland ice sheet: ~2,850,000 km³ (9% of global land ice)
- Mountain glaciers: ~158,000 km³
- Total groundwater: 23,400,000 km³
- Fresh groundwater: ~10,530,000 km³
- Permafrost: ~300,000 km³
- Lakes (freshwater): ~91,000 km³
- Lakes (saline): ~85,400 km³
- Soil moisture: ~16,500 km³
- Atmospheric water: ~12,900 km³
- Wetlands: ~11,500 km³
- Rivers: ~2,120 km³
- Biological water: ~1,120 km³

- Global precipitation: $\sim 505,000 \text{ km}^3/\text{yr}$
- Global evaporation: $\sim 505,000 \text{ km}^3/\text{yr}$
- Global continental runoff: $\sim 36,000 \text{ km}^3/\text{yr}$
- Submarine groundwater discharge: $\sim 2,400 \text{ km}^3/\text{yr}$

Virtual Water — Commodity Footprints (litres per unit)

- 1 kg beef — 15,415 L
- 1 kg sheep/goat — 8,763 L
- 1 kg pork — 5,988 L
- 1 kg chicken — 4,325 L
- 1 kg cheese — 3,178 L
- 1 kg eggs — 3,265 L
- 1 kg rice — 2,497 L
- 1 kg wheat — 1,827 L
- 1 kg maize — 1,222 L
- 1 kg potatoes — 287 L
- 1 kg apples — 822 L
- 1 kg bananas — 790 L
- 1 kg tomatoes — 214 L
- 1 cotton T-shirt — $\sim 2,700 \text{ L}$
- 1 pair jeans — $\sim 7,600 \text{ L}$
- 1 A4 sheet of paper — $\sim 10 \text{ L}$
- 1 glass of milk (250 mL) — $\sim 255 \text{ L}$
- 1 cup coffee (125 mL) — $\sim 140 \text{ L}$
- 1 cup tea (250 mL) — $\sim 30 \text{ L}$
- 1 chocolate bar (100 g) — $\sim 1,700 \text{ L}$
- 1 L beer — $\sim 300 \text{ L}$
- 1 L wine — $\sim 870 \text{ L}$

Source: Hoekstra & Mekonnen, Water Footprint Network (2011, 2020). These figures are global averages; local production footprints vary by climate, irrigation practice and yield.

Freshwater Withdrawal by Sector (global averages)

- Agriculture (overwhelmingly irrigation) — 70% (FAO AQUASTAT)
- Industry (including thermoelectric cooling) — 19%
- Domestic (municipal) — 11%

Sectoral shares vary markedly: agriculture is 82% in low-income countries, 44% in high-income; industry is 40% in Europe but under 5% in sub-Saharan Africa.

Key Countries by Freshwater Availability ($\text{m}^3/\text{person}/\text{yr}$, FAO AQUASTAT)

- Iceland — > 500,000 (highest in Europe)
- Canada — ~81,000
- Norway — ~72,000
- Brazil — ~40,000
- Russia — ~31,000
- USA — ~9,000
- France — ~3,000
- UK — ~2,200
- Germany — ~1,900
- Spain — ~2,400
- India — ~1,100 (stressed)
- China — ~2,000 (above the 1,700 threshold nationally; north China severely stressed)
- South Africa — ~850 (scarcity)
- Israel — ~200 (absolute scarcity; offset by desalination)
- Saudi Arabia — ~75 (absolute scarcity)
- UAE — ~16 (absolute scarcity)
- Kuwait — ~5 (lowest globally)

Key Dates for Eduqas Candidates

- 1922 — Colorado River Compact (over-allocation baseline)
- 1929 — Anglo-Egyptian Nile Waters Agreement
- 1959 — Egypt–Sudan Nile Waters Agreement (55.5 + 18.5 km³)
- 1960 — Indus Waters Treaty (India/Pakistan)
- 1960s — Soviet Aral diversion begins
- 1968–85 — Sahel drought
- 1970s — Ogallala aquifer mining accelerates
- 1986 — Slow Food founded (Italy)
- 1992 — Dublin Principles (water as economic good); UNECE Water Convention
- 1993 — Allan coins “virtual water”
- 1997 — UN Watercourses Convention adopted
- 2000 — EU Water Framework Directive; Cochabamba Water War
- 2001–09 — Millennium Drought (Australia)
- 2005 — Kok-Aral Dam completed (North Aral partial recovery)
- 2010 — UN resolution: human right to water and sanitation
- 2011 — GERD construction begins; Fukushima triggers global nuclear-cooling water review
- 2012 — Murray–Darling Basin Plan; UK Flood and Water Management Act SuDS focus
- 2012–16 — California drought (1,200-yr extreme)
- 2014 — UN Watercourses Convention enters into force
- 2015 — SDGs adopted; SDG 6 on water
- 2018 — Cape Town Day Zero narrowly avoided
- 2019 — Jakarta capital relocation announced
- 2020–23 — GERD reservoir filled without binding agreement
- 2021 — First federal Tier 1 shortage declaration, Colorado
- 2022 — Lake Mead at 27% capacity (record low)
- 2023 — Colorado Basin 3 MAF cut agreement; Klamath dam removal begins
- 2024 — Klamath dam removal completed; GERD fifth filling
- 2025 — GERD officially inaugurated (September)

- 2025 — Mid-decade checkpoint for SDG 6 monitoring (off-track globally)

Named Academics and Frameworks

- **Malin Falkenmark** — water-stress thresholds (1,700 / 1,000 / 500 m³)
- **Tony Allan** — virtual water concept (1993); Stockholm Water Prize 2008
- **Arjen Hoekstra** — water-footprint methodology (co-founder Water Footprint Network)
- **John Hewlett** — variable source area model of storm runoff
- **Robert Horton** — infiltration-excess overland flow theory (1933)
- **IPCC AR6 WG1 (2021)** — hydrological intensification evidence
- **Dublin Principles (1992)** — water as finite & economic good; participatory; gender-aware; ecosystem-based
- **Global Water Partnership** — IWRM framework
- **FAO AQUASTAT** — global freshwater database
- **UN-Water** — inter-agency coordination; SDG 6 Synthesis Reports

18. Glossary — Quick Reference

Term	Definition
Aquiclude	Impermeable rock layer that confines an aquifer
Aquifer	Saturated permeable rock yielding usable water
Baseflow	River flow sustained by groundwater between storms
Clausius–Clapeyron	~7%/°C rise in atmospheric water holding capacity
Cryosphere	Frozen water component: ice sheets, glaciers, permafrost, sea ice
Discharge (Q)	River flow rate, m ³ /s; Q = velocity × cross-sectional area
Eutrophication	Nutrient-driven algal blooms and oxygen depletion

Falkenmark indicator	Stress threshold of 1,700 m ³ /person/year
Field capacity	Maximum water soil holds against gravity
Groundwater recharge	Net addition to an aquifer from percolation or injection
Hortonian overland flow	Infiltration-excess surface runoff
Hydrograph	Graph of discharge vs time
Infiltration capacity	Maximum rate soil can absorb water
IWRM	Integrated Water Resources Management (basin-scale)
Lag time	Gap between peak rainfall and peak discharge
PET / AET	Potential vs actual evapotranspiration
Residence time	Store volume / flux rate
Saturation-excess OLF	Runoff from soils already saturated
SDG 6	UN Sustainable Development Goal on water & sanitation
SuDS	Sustainable Drainage Systems
Transpiration	Water loss via plant stomata
Variable source area	Dynamic saturated contributing area (Hewlett)
Virtual water	Water embedded in traded commodities (Allan)
Water budget	$P = Q + E + \Delta S$
Water stress	<1,700 m ³ /person/year renewable water
Watershed	Topographic boundary of a drainage basin

Key Takeaways

- The global hydrological cycle is a closed system with total volume ~ 1.39 billion km^3 ; 96.5% is saline ocean, and only $\sim 0.001\%$ (0.04% of fresh water) is atmospheric at any moment.
- Residence times range from ~ 10 days (atmosphere) to 10,000+ years (ice sheets), controlling sustainability of abstraction.
- At drainage-basin scale the cycle is open: $P = Q + E + \Delta S$; urbanisation and deforestation produce flashier hydrographs with shorter lag times.
- Climate change intensifies the cycle ($\sim 7\%/^{\circ}\text{C}$ via Clausius–Clapeyron) — more extreme floods and droughts, shifted snowpack, shrinking glaciers.
- Water stress (Falkenmark, $< 1,700 \text{ m}^3/\text{cap}/\text{yr}$) affects growing shares of the world; 1.2 bn in physical scarcity, 1.6 bn in economic scarcity.
- Transboundary basins (~ 310 globally) combine hydrological interdependence with political asymmetry — illustrated by Aral, Colorado, GERD/Nile, Murray–Darling.
- Hard engineering (dams, transfers, desalination) and soft / integrated management (IWRM, reuse, demand management) are complementary, not alternatives.
- Virtual water trade ($\sim 30\%$ of global freshwater) links consumption in MEDCs to abstraction in LEDCs; water footprints reveal hidden dependencies.
- Sustainable water futures require physical-human integration: technical efficiency + pricing + governance reform + dietary change.